

Noise Models

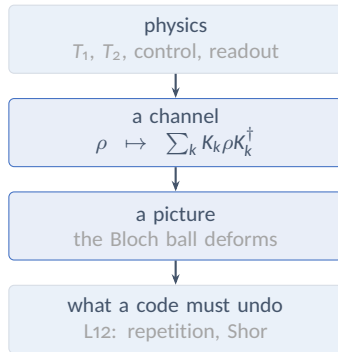
What goes wrong in a qubit, and how we describe it

Lecture 11

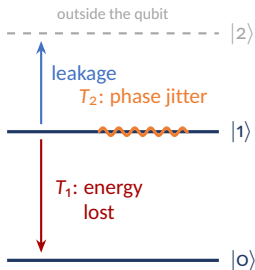
Term 261

Agenda

- 1 Where noise comes from, and how big it is (as of 2026)
- 2 Just enough machinery: density matrices, the Bloch ball, Kraus channels
- 3 Six channels, one Bloch ball each — and all four main ones animated
- 4 Worked fidelities: which states each channel hurts
- 5 Why codes only need to fight X and Z
- 6 Coherent errors, readout, leakage, bursts — and the independence assumption



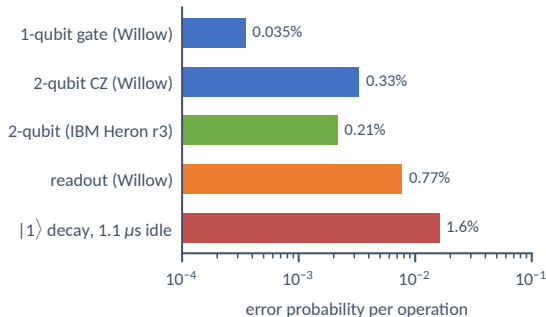
What goes wrong in a real qubit



Source	What it does	Model (this lecture)
Energy relaxation	$ 1\rangle \rightarrow 0\rangle$, time T_1	amplitude damping
Dephasing	random phase, time T_2	phase damping (= flip)
Imperfect control	over/under-rotation	coherent $e^{-i\epsilon X}$
Crosstalk	a gate disturbs a neighbour	correlated 2-qubit errors
Readout error	wrong classical outcome	flipped result, prob. q
Leakage	leaves $\{ 0\rangle, 1\rangle\}$	not a Pauli error at all
Cosmic rays	many qubits fail at once	correlated bursts

The first two are the environment *learning* about the qubit; the next three are *our* mistakes; the last two break the independent-qubit picture itself.

How big are these errors? (as of 2026)



Best published superconducting chips

- T_1 : 68 μ s (Willow) to 350 μ s median (IBM Nighthawk); T_2 similar
- Two-qubit gates: ≈ 0.2 – 0.3% error
- Readout: $\approx 1\%$ — the worst single step
- Surface-code cycle: 1.1 μ s
- Bursts: $\sim$ once an hour, ~ 30 qubits

Takeaway: every operation fails with probability 10^{-3} – 10^{-2} , but factoring RSA-2048 needs $\approx 2.6 \times 10^9$ Toffoli gates. That gap is why we need codes.

Google Quantum AI, Willow spec sheet (Dec 2024); Acharya et al., Nature 638, 920 (2025); IBM product update, 5 Jan 2026; Gidney & Ekerå, Quantum 5, 433 (2021).

One qubit, two kinds of uncertainty

Source A prepares $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$. Source B flips a coin and prepares $|0\rangle$ or $|1\rangle$.

	A: $ +\rangle$	B: coin
measure Z: $P(0)$	0.5	0.5
measure X: $P(+)$	1	0.5
$\langle X \rangle$	1	0

$$\rho_A = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix} \quad \rho_B = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix}$$

The off-diagonals

are the *coherences*: they carry the phase between $|0\rangle$ and $|1\rangle$. Most noise works by eating them.

| B is also a coin flip between $|+\rangle$ and $|-\rangle$: $\frac{1}{2}|+\rangle\langle+| + \frac{1}{2}|-\rangle\langle-| = \frac{1}{2}\mathbb{I}$. Can any experiment tell the two recipes apart? (No — same ρ , same predictions.)

The density matrix

A state $|\psi_i\rangle$ prepared with probability p_i :

$$\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|$$

A legal state

Hermitian	$\rho = \rho^\dagger$
positive	$\langle\varphi \rho \varphi\rangle \geq 0$
unit trace	$\text{tr } \rho = 1$

Pure $\rho = |\psi\rangle\langle\psi| \Leftrightarrow \text{purity } \text{tr } \rho^2 = 1$;
mixed $\Leftrightarrow \text{tr } \rho^2 < 1$ (qubit minimum $\frac{1}{2}$).

Everything we need from it

- Expectation: $\langle O \rangle = \text{tr}(\rho O)$
- Unitary gate: $\rho \mapsto U\rho U^\dagger$
- Fidelity with a pure target:

$$F = \langle\psi|\rho|\psi\rangle$$

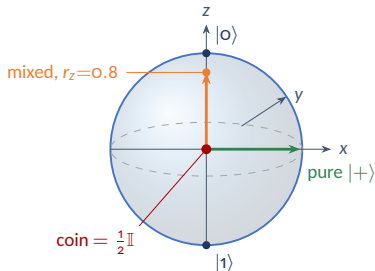
— the probability of passing the test “is it $|\psi\rangle$?”

One more origin of mixedness: a qubit entangled with an environment we cannot see is described by its reduced state $\rho = \text{tr}_E |\Psi\rangle\langle\Psi|_{SE}$. That is how noise arises (next-but-one slide).

The Bloch ball

Every qubit state is

$$\rho = \frac{1}{2}(\mathbb{I} + xX + yY + zZ) = \frac{1}{2}(\mathbb{I} + \mathbf{r} \cdot \boldsymbol{\sigma}), \quad \mathbf{r} = (\text{tr } \rho X, \text{tr } \rho Y, \text{tr } \rho Z), \quad \|\mathbf{r}\| \leq 1.$$



State	$\mathbf{r}$	$\text{tr } \rho^2 = \frac{1}{2}(1 + \ \mathbf{r}\ ^2)$
$ +\rangle$ (A)	$(1, 0, 0)$	1
$0.9 0\rangle\langle 0 + 0.1 1\rangle\langle 1 $	$(0, 0, 0.8)$	0.82
coin (B)	$(0, 0, 0)$	0.5

- **Surface:** pure states (the Bloch sphere). **Inside:** mixed
- **Unitaries rotate** the ball rigidly: $\|\mathbf{r}\|$ is unchanged
- **Noise moves states inward:** the ball shrinks, squashes or slides

Takeaway: a noise model for one qubit is a map of the Bloch ball into itself. For every channel today it is affine, $\mathbf{r} \mapsto M\mathbf{r} + \mathbf{c}$, so the ball becomes an ellipsoid.

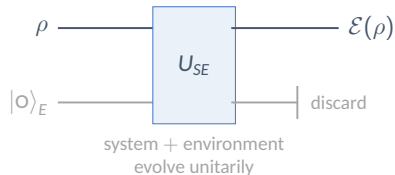
Quantum channels: the Kraus form

Operator-sum (Kraus) representation

$$\mathcal{E}(\rho) = \sum_k K_k \rho K_k^\dagger, \quad \sum_k K_k^\dagger K_k = \mathbb{I}$$

- Completeness $\Leftrightarrow \text{tr } \mathcal{E}(\rho) = 1$; one operator $K_0 = U$ is a gate
- **Reading:** the environment applied K_k with probability $p_k = \text{tr}(K_k \rho K_k^\dagger)$; we were not told which
- If $K_k = \sqrt{p_k} U_k$: “apply U_k with probability p_k ”
- Every physical map of ρ has this form (at most 4 K_k for a qubit); the set is not unique

Nielsen & Chuang (2010), §8.2.3–8.2.4.



$$\mathcal{E}(\rho) = \text{tr}_E [U_{SE}(\rho \otimes |o\rangle\langle o|_E)U_{SE}^\dagger]$$

$$\text{with } K_k = {}_E\langle k| U_{SE} |o\rangle_E.$$

Bit flip

Story. With probability p the qubit is hit by an X : $|0\rangle \leftrightarrow |1\rangle$. The quantum version of the classical binary symmetric channel.

Kraus: $K_0 = \sqrt{1-p}\mathbb{I}$, $K_1 = \sqrt{p}X$

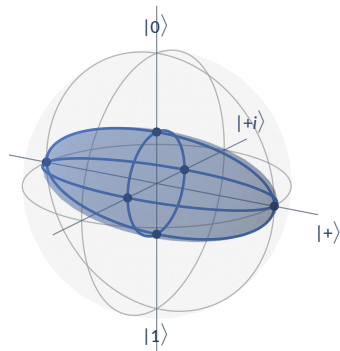
Action: $\mathcal{E}(\rho) = (1-p)\rho + pX\rho X$

$$\begin{pmatrix} a & b \\ b^* & d \end{pmatrix} \mapsto \begin{pmatrix} (1-p)a + pd & (1-p)b + pb^* \\ (1-p)b^* + pb & (1-p)d + pa \end{pmatrix}$$

Bloch: $(x, y, z) \mapsto (x, (1-2p)y, (1-2p)z)$

Who gets hurt

$|0\rangle, |1\rangle$: $F = 1 - p$. $|\pm\rangle$ lie on the x axis and are **untouched** ($X|\pm\rangle = \pm|\pm\rangle$).



$p = 0.3$: the ball is squeezed onto the x axis

Phase flip

Story. With probability p the qubit is hit by a Z : $|+\rangle \leftrightarrow |-\rangle$. Populations survive, the relative phase does not. **No classical analogue.**

Kraus: $K_0 = \sqrt{1-p} \mathbb{I}$, $K_1 = \sqrt{p} Z$

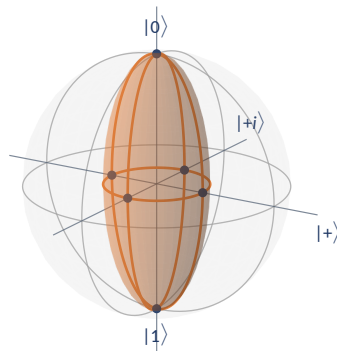
Action: $\mathcal{E}(\rho) = (1-p)\rho + pZ\rho Z$

$$\begin{pmatrix} a & b \\ b^* & d \end{pmatrix} \mapsto \begin{pmatrix} a & (1-2p)b \\ (1-2p)b^* & d \end{pmatrix}$$

Bloch: $(x, y, z) \mapsto ((1-2p)x, (1-2p)y, z)$

Who gets hurt

$|\pm\rangle, |\pm i\rangle$: $F = 1-p$. $|0\rangle, |1\rangle$ are **untouched** — a classical bit stored in $|0\rangle / |1\rangle$ never notices.



$p = 0.3$: the ball is squeezed onto the z axis

Takeaway: this is the error a classical repetition code cannot see. L12 fixes it by repeating in the $|\pm\rangle$ basis.

Bit-phase flip

Story. With probability p the qubit is hit by $Y = iXZ$: a bit flip *and* a phase flip at once (the global phase i is invisible).

Kraus: $K_0 = \sqrt{1-p} \mathbb{I}$, $K_1 = \sqrt{p} Y$

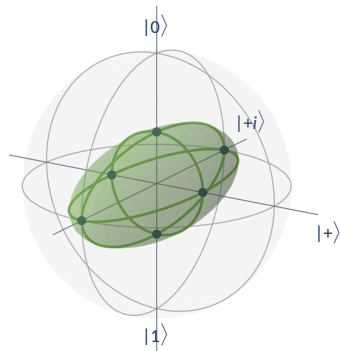
Action: $\mathcal{E}(\rho) = (1-p)\rho + pY\rho Y$

Bloch: $(x, y, z) \mapsto ((1-2p)x, y, (1-2p)z)$

Who gets hurt

$|0\rangle, |1\rangle, |\pm\rangle$: $F = 1 - p$. Only $|\pm i\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm i|1\rangle)$, on the y axis, are untouched.

Pattern of the three flips: **Pauli P leaves its own axis alone and shrinks the other two by $1 - 2p$.**



$p = 0.3$: the ball is squeezed onto the y axis

Depolarizing

Story. With probability p one of X, Y, Z hits, each equally likely. No preferred direction: the standard benchmark for codes and thresholds.

Our convention: $\mathcal{E}(\rho) = (1 - p)\rho + \frac{p}{3}(X\rho X + Y\rho Y + Z\rho Z)$

Kraus: $\sqrt{1-p}\mathbb{I}, \sqrt{p/3}X, \sqrt{p/3}Y, \sqrt{p/3}Z$

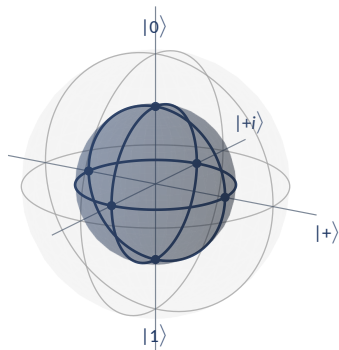
Bloch: $\mathbf{r} \mapsto (1 - \frac{4p}{3})\mathbf{r}$ — a uniform shrink

The other parametrization

Since $\rho + X\rho X + Y\rho Y + Z\rho Z = 2\mathbb{I}$: $\mathcal{E}(\rho) = (1 - q)\rho + q\frac{\mathbb{I}}{2}$ with $q = \frac{4p}{3}$. “Replace by a random state with probability q .” Check which p a paper means!

Every pure state: $F = 1 - \frac{2p}{3}$. At $p = \frac{3}{4}$ the output is $\frac{1}{2}\mathbb{I}$.

Both forms: Nielsen & Chuang (2010), §8.3.4.



$p = 0.3$: a smaller sphere, radius $1 - \frac{4p}{3} = 0.6$

Amplitude damping (T_1)

Story. The excited state $|1\rangle$ gives its energy to the environment and falls to $|0\rangle$.

Kraus: $K_0 = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-\gamma} \end{pmatrix}$ (no jump), $K_1 = \begin{pmatrix} 0 & \sqrt{\gamma} \\ 0 & 0 \end{pmatrix} = \sqrt{\gamma} |0\rangle\langle 1|$ (jump)

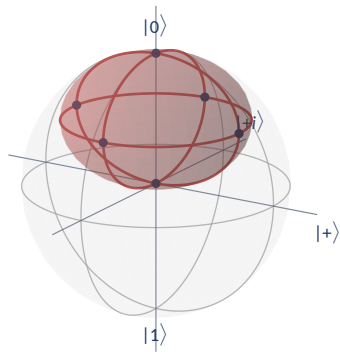
$$\begin{pmatrix} a & b \\ b^* & d \end{pmatrix} \mapsto \begin{pmatrix} a + \gamma d & \sqrt{1-\gamma} b \\ \sqrt{1-\gamma} b^* & (1-\gamma) d \end{pmatrix}$$

Bloch: $(x, y, z) \mapsto (\sqrt{1-\gamma} x, \sqrt{1-\gamma} y, \gamma + (1-\gamma)z)$

Time: $\gamma = 1 - e^{-t/T_1}$ (two waits compose: $1-\gamma$ multiplies)

Not like the others

Not a Pauli channel, and not *unital*: the centre moves to $(0, 0, \gamma)$. Only $|0\rangle$ is untouched; $|1\rangle$ is hurt most. Even “no jump” changes the state.



$\gamma = 0.5$: shrinks and slides toward $|0\rangle$

Phase damping (T_2)

Story. No energy changes hands; the qubit frequency jitters and the relative phase is lost.

Kraus: $K_0 = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-\lambda} \end{pmatrix}$, $K_1 = \begin{pmatrix} 0 & 0 \\ 0 & \sqrt{\lambda} \end{pmatrix}$

$$\begin{pmatrix} a & b \\ b^* & d \end{pmatrix} \mapsto \begin{pmatrix} a & \sqrt{1-\lambda} b \\ \sqrt{1-\lambda} b^* & d \end{pmatrix}$$

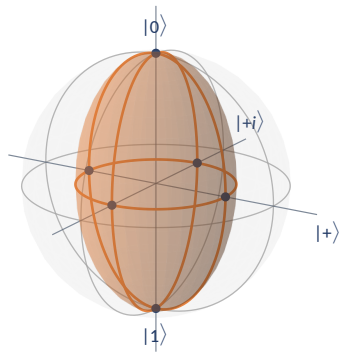
Bloch: $(x, y, z) \mapsto (\sqrt{1-\lambda} x, \sqrt{1-\lambda} y, z)$

Same channel as the phase flip

Phase damping $\lambda =$ phase flip with $p = \frac{1}{2}(1 - \sqrt{1-\lambda})$ (e.g. $\lambda = 0.36 \leftrightarrow p = 0.1$). Different Kraus sets, identical map.

With T_1 : coherence $\propto e^{-t/T_2}$, $\frac{1}{T_2} = \frac{1}{2T_1} + \frac{1}{T_\phi}$, so $T_2 \leq 2T_1$.

Nielsen & Chuang (2010), §8.3.5–8.3.6; T_2 relation: Krantz et al. (2019), §II.

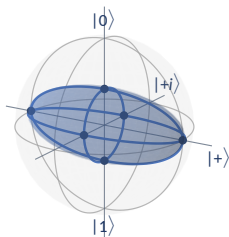


$\lambda = 0.64$: the same ball as phase flip $p = 0.2$

Four channels, one picture

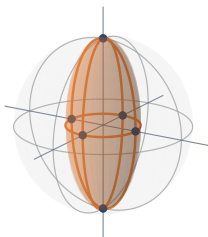
$p = 0.29$

Bit flip



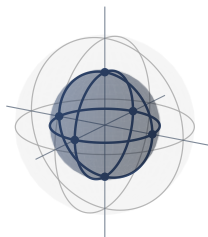
$$(x, (1-2p)y, (1-2p)z)$$

Phase flip



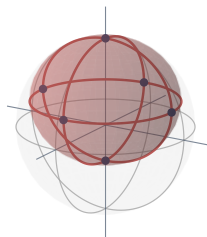
$$((1-2p)x, (1-2p)y, z)$$

Depolarizing



$$(1-\frac{4p}{3})\vec{r}$$

Amplitude damping



$$\gamma = p: \sqrt{1-\gamma} (x, y), z \rightarrow \gamma + (1-\gamma)z$$

Unital (centre fixed): bit flip, phase flip and depolarizing only *shrink* — toward an axis, or uniformly. The dots are the images of $|0\rangle, |1\rangle, |\pm\rangle, |\pm i\rangle$.

Non-unital: amplitude damping also *slides* the ball toward $|0\rangle$. At $p = \frac{1}{2}$ the flips crush the ball to a line segment; depolarizing leaves a sphere of radius $\frac{1}{3}$.

Worked example: which states does each channel hurt?

Fidelity $F = \langle \psi | \mathcal{E}(|\psi\rangle\langle\psi|) | \psi \rangle$ at strength 0.1 (p , γ or λ):

Channel	$ 0\rangle$	$ 1\rangle$	$ +\rangle$	$ +i\rangle$
bit flip	0.9	0.9	1	0.9
phase flip	1	1	0.9	0.9
bit-phase flip	0.9	0.9	0.9	1
depolarizing	0.933	0.933	0.933	0.933
amplitude damp.	1	0.9	0.974	0.974
phase damping	1	1	0.974	0.974

Two by hand

Bit flip on $|0\rangle$: $\mathcal{E} = 0.9 |0\rangle\langle 0| + 0.1 |1\rangle\langle 1|$,
 $F = 0.9$

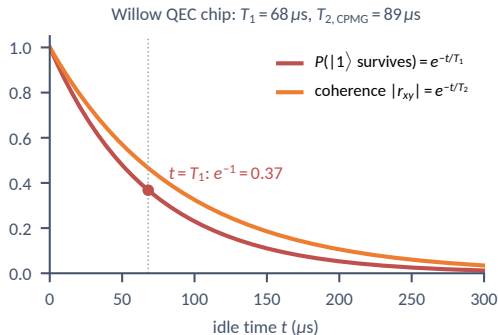
Bit flip on $|+\rangle$: $X|+\rangle = |+\rangle$, so $\mathcal{E} = |+\rangle\langle +|$,
 $F = 1$

Depolarizing:

$F = \frac{1}{2} \left(1 + \left(1 - \frac{4p}{3} \right) \right) = 1 - \frac{2p}{3} = 0.933$ for
every pure state

Takeaway: a code must protect *all* states, so it must handle the worst column of every row — both X-type and Z-type damage.

Worked example: a qubit that waits



Idle for $t = T_1$ (amplitude damping only):

$$\gamma = 1 - e^{-1} = 0.632$$

- $|1\rangle$: $P(\text{still } |1\rangle) = 1 - \gamma = e^{-1} = 0.37$
- $|+\rangle$: $\mathbf{r} = (e^{-1/2}, 0, 1 - e^{-1}) = (0.61, 0, 0.63)$,
 $F = \frac{1}{2}(1 + 0.61) = 0.80$

Idle for one QEC cycle (Willow: $1.1 \mu\text{s}$, $T_1 = 68 \mu\text{s}$)

- $\gamma = 1 - e^{-1.1/68} = 0.016$ per cycle for $|1\rangle$
- about 62 cycles per T_1

Takeaway: even *doing nothing* is an error channel. Codes must measure syndromes fast compared with T_1 and T_2 — and this sets the per-cycle p in L13 and L16.

$T_1, T_{2,\text{CPMG}}$, cycle time: Acharya et al., Nature 638, 920 (2025).

Every error is a combination of Paulis

Any 2×2 matrix — unitary or not — expands in the Pauli basis:

$$E = e_0 \mathbb{I} + e_1 X + e_2 Y + e_3 Z, \quad e_j = \frac{1}{2} \text{tr}(P_j^\dagger E).$$

Error operator	Pauli expansion	From
$ 0\rangle\langle 0 $	$\frac{1}{2}(\mathbb{I} + Z)$	T251 notes
$ 0\rangle\langle 1 $	$\frac{1}{2}(X + iY)$	T_1 jump $K_1/\sqrt{\gamma}$
$e^{-i\epsilon X}$	$\cos \epsilon \mathbb{I} - i \sin \epsilon X$	over-rotation

And $Y = iXZ$

So every single-qubit error is a combination of

$$\mathbb{I}, \quad X, \quad Z, \quad XZ.$$

- A code that corrects X and Z on a qubit (hence XZ) corrects *any* E on that qubit: syndrome measurement collapses the superposition onto one Pauli term
- This is the **discretization of errors**. L12 sees it happen in the Shor code; L13 proves it (Knill–Laflamme)

Pauli channels and twirling

Pauli channel: $\mathcal{E}(\rho) = p_I \rho + p_X X\rho X + p_Y Y\rho Y + p_Z Z\rho Z$ — all of today's channels except amplitude damping.

Twirling turns any channel into one: sandwich it between a random Pauli P and P again,

$$\mathcal{T}(\mathcal{E})(\rho) = \frac{1}{4} \sum_{P \in \{I, X, Y, Z\}} P \mathcal{E}(P\rho P) P = \sum_P p_P P\rho P,$$

with $p_P = \sum_k |c_{k,P}|^2$, where $K_k = \sum_P c_{k,P} P$. Cross terms between different Paulis average to zero.

In hardware: **randomized compiling** inserts random Paulis around gates and compiles them away — same circuit, stochastic noise.

Takeaway: Pauli channels are not just convenient: with twirling they are what the code actually sees.

Two examples

Amplitude damping, $\gamma = 0.1$, twirled:

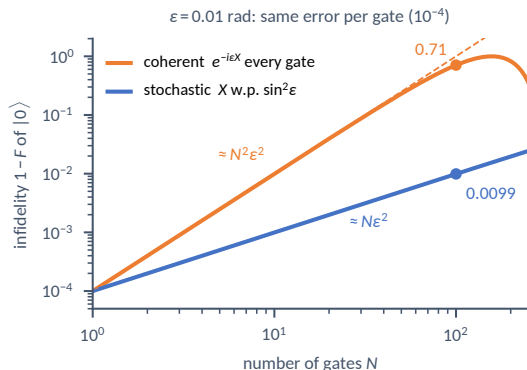
p_I	p_X	p_Y	p_Z
0.9493	0.025	0.025	0.0007

$$p_X = p_Y = \frac{\gamma}{4}, \quad p_Z = \frac{1}{4}(1 - \sqrt{1 - \gamma})^2$$

Over-rotation $e^{-i\epsilon X}$, twirled: a bit flip with $p = \sin^2 \epsilon$ (10^{-4} at $\epsilon = 0.01$)

—
Bennett et al., PRA 54, 3824 (1996); Wallman & Emerson, PRA 94, 052325 (2016).

Coherent versus stochastic errors



A gate over-rotates by $\epsilon = 0.01$ rad every time.
Start in $|0\rangle$, apply it N times.

Coherent: $e^{-iN\epsilon X} |0\rangle$ — *amplitudes* add,

$$1 - F = \sin^2(N\epsilon) \approx N^2 \epsilon^2$$

Stochastic (twirled): X with prob. $\sin^2 \epsilon$ — *probabilities* add,

$$1 - F = \frac{1}{2} (1 - (1 - 2 \sin^2 \epsilon)^N) \approx N \epsilon^2$$

At $N = 100$: **0.71** versus **0.0099** — $72\times$ worse.

Takeaway: same error per gate (10^{-4}), very different damage. Systematic errors are calibrated out or randomized into stochastic ones before a code ever sees them.

Other noise QEC must face

Noise	Model	Why it matters for codes
Readout	classical flip of the outcome with prob. q ($P(1 0) = q$) = an X just before an ideal Z measurement; asymmetric in practice	syndromes are themselves wrong ($q \approx 0.8\%$ on Willow) $\Rightarrow$ repeat syndrome measurements (L16)
Leakage	population leaves $\{ 0\rangle, 1\rangle\}$ (e.g. to $ 2\rangle$)	invisible to Pauli checks, spreads through 2-qubit gates; Willow runs leakage removal every cycle
Crosstalk	two-qubit Pauli channels $\sum p_{PQ} (P \otimes Q) \rho (P \otimes Q)$, always-on ZZ coupling	errors on neighbours are <i>correlated</i>
Bursts	cosmic rays / radioactivity: many qubits fail together	about once an hour over ~ 30 qubits on Willow; defeats any fixed distance
Non-Markovian	noise with memory: drift, $1/f$ noise, the channel at step t depends on earlier steps	errors correlated in <i>time</i> ; recalibration, decoupling

McEwen et al., Nature Physics 18, 107 (2022); Acharya et al., Nature 638, 920 (2025).

The assumption every code analysis makes

Independent, identically distributed noise

Each qubit, at each step, suffers its own Pauli channel with total error probability p , independently of all others. Then

$$P(k \text{ errors on } n \text{ qubits}) = \binom{n}{k} p^k (1-p)^{n-k}.$$

Small p makes *many* errors rare — this is what a distance- d code exploits (L13).

The running example: 3 qubits, $p = 0.01$, fail at ≥ 2 flips, $P = 3p^2 - 2p^3$ if independent:

Noise	$P(\text{fail})$
independent	2.98×10^{-4}
+ burst w.p. 10^{-4}	3.98×10^{-4}
+ burst w.p. 10^{-3}	1.30×10^{-3}

A burst that flips all three at once, a thousand times rarer than p , still makes the code $4.4\times$ worse.

The catch: codes are designed for independent noise. Correlated errors must be engineered away (shielding, gap engineering, leakage removal) — the code cannot do it alone.

Summary

Channel	Kraus operators	Bloch map	Physical origin	Hurts
Bit flip	$\sqrt{1-p}\mathbb{I}, \sqrt{p}X$	$(x, (1-2p)y, (1-2p)z)$	stray X rotations	all but $ \pm\rangle$
Phase flip	$\sqrt{1-p}\mathbb{I}, \sqrt{p}Z$	$((1-2p)x, (1-2p)y, z)$	frequency noise	all but $ 0\rangle, 1\rangle$
Bit-phase	$\sqrt{1-p}\mathbb{I}, \sqrt{p}Y$	$((1-2p)x, y, (1-2p)z)$	both at once	all but $ \pm i\rangle$
Depolarizing	$\sqrt{1-p}\mathbb{I}, \sqrt{p/3}X, Y, Z$	$(1 - \frac{4p}{3})\mathbf{r}$	benchmark model	all equally
Amp. damping	$\begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-\gamma} \end{pmatrix}, \begin{pmatrix} 0 & \sqrt{\gamma} \\ 0 & 0 \end{pmatrix}$	$(\sqrt{1-\gamma}x, \sqrt{1-\gamma}y, \gamma + (1-\gamma)z)$	energy loss, T_1	all but $ 0\rangle$
Phase damping	$\begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-\lambda} \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & \sqrt{\lambda} \end{pmatrix}$	$(\sqrt{1-\lambda}x, \sqrt{1-\lambda}y, z)$	dephasing, T_2	all but $ 0\rangle, 1\rangle$

- Noise = a channel $\rho \mapsto \sum_k K_k \rho K_k^\dagger$ = a deformation of the Bloch ball
- Every error expands in $\mathbb{I}, X, Y, Z$, and $Y = iXZ$; twirling makes the noise a Pauli channel
- Coherent errors grow as $N^2\epsilon^2$, stochastic ones as $N\epsilon^2$
- Codes assume independent noise; readout, leakage and bursts must be handled on top

References

Textbooks and reviews

- M. A. Nielsen, I. L. Chuang (2010). *Quantum Computation and Quantum Information*, 10th anniv. ed. Cambridge Univ. Press — §2.4 (density operator), ch. 8 (quantum noise and operations).
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Every Bloch map, fidelity and twirl on these slides was computed by applying the Kraus operators to density matrices (scripts `L11_channels_verify.py`, `L11_bloch_figs.py`, `L11_plots.py`).